

Lossless Assembly of an Eight-Point Certificate for Simple Critical Zeros of the Riemann Zeta Function

RH–Weil research project

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Abstract

We combine an existing seven-gap certificate for a perturbed Montgomery–Taylor window with the close-pair separation mechanism used by Knausgård. An explicit band-limited majorant has mass strictly below two and dominates the same perturbed window. It removes the spectral truncation loss on the separated set, while disjoint close pairs pay the same per-point reward. Sliding the existing eight-point certificate directly over the separated set gives

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T)}{N(T)} \geq \frac{66812491}{99194997} = 0.6735469834229644 \dots$$

Here the denominator counts all nontrivial zeros with multiplicity, and the numerator counts simple zeros on the critical line. The analytic argument is unconditional and retains off-line zeros. The computational input is the previously admitted PC8CL continuous seven-gap certificate. We give a separate, small exact rational certificate for the new majorant and the energy constant. This paper does not claim a complete Lean kernel verification or external peer-review certification.

1 Statement, provenance and comparison

Let $N(T)$ count the nontrivial zeros ρ of ζ , with $0 < \Im \rho \leq T$, including multiplicity. Let $N_0^s(T)$ count the simple zeros among them with $\Re \rho = 1/2$.

Theorem 1.1. *Using the continuous PC8CL certificate stated in Input 2.1, one has, unconditionally,*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T)}{N(T)} \geq \frac{66812491}{99194997}. \quad (1)$$

No zero-free strip, Riemann hypothesis or higher correlation conjecture is assumed.

The local certificate and the window below are existing inputs from the fixed public AM eight-point construction [4]. Their prior square-root spectral assembly gave the admitted rational bound $118513839290/175971686899$. The present change is in the assembly: we use a band-limited majorant, remove disjoint close pairs, and apply the old local inequality without a square-root loss. We adapt the simple-point separation mechanism in Knausgård [5, Section 5], whose v1 uses a different window and local certificate to obtain $1669159/2478195$. We reprove the finite and analytic transfer needed here. Our exact comparison with that stated v1 benchmark is

$$\frac{66812491}{99194997} - \frac{1669159}{2478195} = \frac{719712074}{81941515196805} > 0. \quad (2)$$

The increase is approximately 0.0008783241 percentage points. This comparison is with a checked, fixed benchmark; it does not establish exhaustive priority over all unpublished or concurrent work. The proportion in (1) is a zero-counting lower bound, rather than a measure of progress toward a proof of RH.

2 Window and admitted local certificate

Put $\beta = 1/\sqrt{2}$ and $Z_0 = \sqrt{2} \sin \beta$. On $I = [-1/2, 1/2]$ define

$$f(u) = \frac{\cos(\sqrt{2}u) + \sum_{j=1}^{12} c_j \cos(2\pi j u)}{Z_0}, \quad K(t) = \int_I f(u) e^{-2\pi i t u} du. \quad (3)$$

Extend f by zero outside I . The integer numerators $10^9 c_j$ are

$$(12310798, -15041681, 3867664, 6489926, -992327, 5580097, \\ -6846472, 3781297, -5670353, 3355089, -450523, -218483).$$

Thus $S = \sum_j |c_j| = 6460471/10^8 < 13/200$. Since $\cos(\sqrt{2}u) \geq 1 - u^2 \geq 3/4$ on I , the numerator is positive. The integer cosine modes integrate to zero, so f is an even probability density. Also

$$Z_0 = \int_I \cos(\sqrt{2}u) du \geq 11/12. \quad (4)$$

For eight ordered points $y_0, \dots, y_7$, write $g_r = y_{r+1} - y_r \geq 0$, $0 \leq r \leq 6$. Set

$$c = \frac{805003}{10^8}, \quad B = \sum_r b_r = \frac{404350}{10^8},$$

$$(10^8 b_r)_{r=0}^6 = (28898, 57272, 75526, 80958, 75526, 57272, 28898).$$

All pair weights a_{ij} are nonnegative. Table 1 gives their integer numerators.

$i \setminus j$	1	2	3	4	5	6	7
0	13630830	0	19565904	20317441	45030671	100000000	200000000
1		29997996	30621878	47766489	79682558	109938656	100000000
2			37356140	69378121	65335210	79682558	45030671
3				38030065	69378121	47766489	20317441
4					37356140	30621878	19565904
5						29997996	0
6							13630830

Table 1: The weights $10^8 a_{ij}$ of the fixed AM certificate.

Computational input 2.1 (Fixed PC8CL certificate). *For every $(g_0, \dots, g_6) \in [0, \infty)^7$,*

$$\sum_{r=0}^6 b_r g_r + \sum_{0 \leq i < j \leq 7} a_{ij} K \left(\sum_{r=i}^{j-1} g_r \right)^2 \geq c. \quad (5)$$

This is a continuous real-gap assertion, not a grid sampling claim. Its provenance, finite replay and continuous soundness evidence are specified in Section 6. We use the complete admitted certificate, rather than the upstream final zeta theorem. Direct integer summation of Table 1 gives

$$10^8 \left(\sum_{i=0}^{7-s} a_{i,i+s} \right)_{s=1}^7 = (199999997, 199999998, 199999996, 199999998, \\ 199999998, 200000000, 200000000). \quad (6)$$

In particular each span-length budget is at most two.

Lemma 2.2 (Direct sliding). *For any m ordered real points $y_0 \leq \dots \leq y_{m-1}$, with span taken to be zero when $m \leq 1$,*

$$\sum_{i \neq j} K(y_i - y_j)^2 \geq c(m-7) - B \text{ span}(y). \quad (7)$$

Proof. For $m \geq 8$, sum (5) over the $m-7$ consecutive eight-point windows. A fixed pair of index distance $s \leq 7$ receives total weight at most $\sum_i a_{i,i+s} \leq 2$; pairs farther apart receive no weight. A fixed adjacent gap receives total coefficient at most $\sum_r b_r = B$. All terms are nonnegative, so the weighted pair sum is at most the full ordered pair sum in (7). This proves the assertion. For $m \leq 7$ its right side is nonpositive. $\square$

3 A band-limited majorant for the same window

We use $\text{sinc } t = \sin(\pi t)/(\pi t)$, with $\text{sinc } 0 = 1$, and put

$$\theta = \frac{4}{5}, \quad \gamma = \frac{61}{100}, \quad g(u) = \gamma \{\text{sinc}(\theta(u-1/2)) + \text{sinc}(\theta(u+1/2))\}^2. \quad (8)$$

Lemma 3.1. *The nonnegative function g satisfies $g \geq f$ on $\mathbb{R}$, $\hat{g}(t) = 0$ for $|t| \geq \theta$, and*

$$\Lambda := \int_{\mathbb{R}} g(u) du = \frac{2\gamma}{\theta} (1 + \text{sinc } \theta) \leq \frac{61}{32} < 2. \quad (9)$$

Proof. Each shifted sinc has a rectangular Fourier transform supported in $[-\theta/2, \theta/2]$. The square in (8) therefore has Fourier support $[-\theta, \theta]$. The convolution is continuous and zero at both boundary points, so the conclusion includes equality $|t| = \theta$. Parseval gives the displayed formula for Λ . As $\sin(4\pi/5) = \sin(\pi/5) \leq \pi/5$, $\text{sinc}(4/5) \leq 1/4$, proving its stated upper bound.

The remaining assertion $g \geq f$ is proved by the small exact rational certificate in Appendix A. Specifically it establishes

$$g(k/256) - f(k/256) > 9/100 \quad (0 \leq k \leq 128), \quad \sup_{u \in I} |(g-f)'(u)| < 16.$$

Both functions are even, and every point of $[0, 1/2]$ is at distance at most $1/512$ from one of these nodes. Hence on I

$$g(u) - f(u) > 9/100 - 16/512 = 47/800 > 0.$$

Outside I the assertion follows from $f = 0$ and $g \geq 0$. The derivative estimate is the essential continuous bridge; the grid inequalities alone would not suffice. $\square$

Lemma 3.2 (Close-pair reward). *For $|t| \leq 4/5$, one has $K(t) > 134/825$ and $K(t)^2 > 1/40 > c$.*

Proof. For $0 \leq t \leq 4/5$, the functions $\cos(\sqrt{2}u)$ and $\cos(2\pi tu)$ both decrease on $[0, 1/2]$. Chebyshev's integral covariance inequality, divided by Z_0 , gives for the unperturbed kernel $K_0(t) \geq \text{sinc } t \geq \text{sinc}(4/5)$. The last monotonicity follows by differentiating $\sin x/x$ on $[0, \pi]$. Using $\pi^2 < 10$ and $\sin x \geq x - x^3/6$ at $x = \pi/5$ gives $\text{sinc}(4/5) > 7/30$. The perturbation of the normalized kernel has modulus at most $S/Z_0 < 39/550$. Thus $K(t) > 7/30 - 39/550 = 134/825$. Finally $(134/825)^2 = 17956/680625 > 1/40 > c$. Evenness covers negative t . $\square$

4 Lossless spectral assembly

Choose even $\chi_\varepsilon \in C_c^\infty((-1/2, 1/2))$, $0 \leq \chi_\varepsilon \leq 1$, tending to one on the interior, and set

$$m_\varepsilon = \int_I f \chi_\varepsilon^2, \quad f_\varepsilon = f \chi_\varepsilon^2 / m_\varepsilon, \quad K_\varepsilon = \widehat{f_\varepsilon}, \quad d_\varepsilon = \|f_\varepsilon - f\|_1.$$

Then $m_\varepsilon \rightarrow 1$, $d_\varepsilon \rightarrow 0$, and $f_\varepsilon \rightarrow f$ in $L^1 \cap L^2$. The square root $\eta_\varepsilon = \sqrt{f_\varepsilon}$ is smooth, even, compactly supported in the interior of I . On the real axis, $|K_\varepsilon - K| \leq d_\varepsilon$ and both kernels have modulus at most one. Throughout the argument fix ε sufficiently small that $m_\varepsilon > 61/64$.

For a finite set Y of n real points let $U = (K_\varepsilon(y_i - y_j))_{i,j}$ be its positive Gram matrix, with unit diagonal. Define

$$\phi_2(t) = t^2 - (t-2)_+^2, \quad J(U) = \text{tr } \phi_2(U) - n.$$

The scalar function ϕ_2 is convex on $[0, \infty)$.

Proposition 4.1. *For every finite Y ,*

$$J(U) \geq cn - B \text{span}(Y) - 7c - 42d_\varepsilon n. \quad (10)$$

Proof. Select a maximal collection of disjoint pairs with distance strictly less than θ . The remaining set S is θ -separated. For any coefficient vector (z_j) on S , the pointwise majorant gives

$$\int f_\varepsilon(u) \left| \sum_{j \in S} z_j e^{-2\pi i y_j u} \right|^2 du \leq \frac{1}{m_\varepsilon} \int g(u) \left| \sum_{j \in S} z_j e^{-2\pi i y_j u} \right|^2 du = \frac{\Lambda}{m_\varepsilon} \sum_{j \in S} |z_j|^2.$$

The cross terms vanish even at separation exactly θ . Thus $0 \preceq U_{SS} \preceq (\Lambda/m_\varepsilon)I \prec 2I$. Normalization by m_ε is necessary here. Consequently $\phi_2(U_{SS}) = U_{SS}^2$.

Write $m = |S|$. Apply each local certificate to K and transfer only its finitely many weighted pair terms to K_ε . Each square changes by at most $3d_\varepsilon$; the total weight in one window is at most 14 by (6). The sliding argument, including the cases $m \leq 7$, yields

$$\text{tr } \phi_2(U_{SS}) - m \geq cm - B \text{span}(S) - 7c - 42d_\varepsilon m.$$

Each removed pair has a Gram matrix with eigenvalues $1 \pm |K_\varepsilon(t)|$ in $[0, 2]$. Its contribution to J is $2K_\varepsilon(t)^2 \geq 2c - 6d_\varepsilon$, by Lemma 3.2.

Diagonalize the principal block for S and each two-point block. Scalar Jensen in the resulting orthonormal basis proves $\text{tr } \phi_2(U) \geq \sum_{\text{blocks}} \text{tr } \phi_2(U_{\text{block}})$. This requires scalar convexity, not operator convexity. Summing the pair rewards and the separated-set reward gives (10), since $\text{span}(S) \leq \text{span}(Y)$ and the pair error is at most $3d_\varepsilon(n - m) \leq 42d_\varepsilon(n - m)$. Empty or small separated sets are covered by the same single boundary charge $7c$. $\square$

5 All-zero Hermitian counting and the energy constant

5.1 The unconditional analytic input

For each fixed ε , the unconditional correlation formula of Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [1, Lemma 5], with the correction described in [2, Section 3], gives

$$\sum_{0 < \Im \rho, \Im \rho' \leq T} K_\varepsilon \left(i(\rho - \rho') \frac{\log T}{2\pi} \right)^2 = (C(f_\varepsilon) + o_\varepsilon(1))N(T), \quad (11)$$

where

$$C(h) = \int_I h(u)^2 du + \int_I \int_I |u - v| h(u) h(v) du dv. \quad (12)$$

The sum includes multiplicities and all off-line zeros. For completeness, the weighted lemma applied to a fixed real even test q supported in $[-1, 1]$ has weight $w(\rho - \rho') = 4/(4 - (\rho - \rho')^2)$ and main term $q(0) + 2 \int_0^1 uq(u) du$. Apply it separately to the two fixed smooth functions $q = f_\varepsilon * f_\varepsilon$ and q'' . For $z = i(\rho - \rho') \log T/(2\pi)$, integration by parts gives

$$\left(\widehat{q}(z) - \frac{\widehat{q''}(z)}{4(\log T)^2} \right) w(\rho - \rho') = K_\varepsilon(z)^2.$$

The second fixed-test contribution is $O_\varepsilon(N/(\log T)^2)$; the first gives (11). The lemma allows signed tests, so its use for q'' is legitimate. This is the fixed-smoothing argument also used in [3]. No estimate uniform in a T -dependent smoothing parameter is used.

5.2 Retaining off-line zeros

Put $L = \log T$ and $z_\rho = -i(\rho - 1/2)L/(2\pi)$. The multiset of these points is conjugation-invariant. First in the complex Hilbert space $L^2(I; \mathbb{C})$ set $v_z(u) = \eta_\varepsilon(u)e^{-2\pi izu}$. For each distinct nonreal conjugate pair write $g_z = (v_z + v_{\bar{z}})/2$ and $h_z = (v_z - v_{\bar{z}})/(2i)$. The vectors v_x for real x , together with g_z and h_z , belong to the real Hilbert space $\mathcal{H}_\mathbb{R} = \{v : \overline{v(u)} = v(-u)\}$: the conjugate reflection J satisfies $Jv_z = v_{\bar{z}}$. Define the finite self-adjoint operator on their real span by

$$A_T = \sum_{\text{distinct real } x} \mu_x v_x \otimes v_x + 2 \sum_{\text{distinct nonreal pairs}} \mu_z (g_z \otimes g_z - h_z \otimes h_z).$$

It acts on the finite span of these real-type vectors. The identity $\|g_z\|^2 - \|h_z\|^2 = 1$ and direct tensor expansion give

$$\text{tr } A_T = N(T), \quad \text{tr } A_T^2 = (C(f_\varepsilon) + o_\varepsilon(1))N(T). \quad (13)$$

Indeed the Hilbert–Schmidt square expands to $\sum_{z,w} K_\varepsilon(z - \bar{w})^2$; conjugation reindexes this as the sum in (11). Individual complex terms need not be positive.

Let P be the sum of the unit features at simple real points, $n = N_0^s(T)$, and $Q = A_T - P$. If r is the number of repeated real points and k the number of nonreal conjugate pairs, then

$$n_+(Q) \leq b := r + k, \quad N(T) \geq n + 2b.$$

Each remaining real point has multiplicity at least two; each nonreal pair contributes at least two zeros and at most one positive rank direction. No location-based deletion has occurred.

Lemma 5.1 (Finite spectral counting). *If $A = P + Q$, P is the sum of n unit features with Gram U , $n_+(Q) \leq b$, $\text{tr } A = N$ and $N \geq n + 2b$, then*

$$\text{tr } A^2 \geq 2(N - n) + \text{tr } \phi_2(U) = 2N - n + J(U). \quad (14)$$

Proof. Enlarge the finite ambient space by zero directions if necessary. If $\lambda_i(P) = p_i$ are its n Gram eigenvalues, including zeros, minmax gives $\lambda_{b+i}(A) \leq p_i$. For $t \leq p$, $p \geq 0$, $4t - t^2 \leq 4p - \phi_2(p)$; also $4t - t^2 \leq 4$ for all real t . After the first b eigenvalues use the first inequality; any remaining eigenvalues are nonpositive and contribute at most zero. Since $\sum p_i = n$, this proves $4N - \text{tr } A^2 \leq 4b + 4n - \text{tr } \phi_2(U)$. Now $4b \leq 2(N - n)$ gives (14). Zeros in the n -dimensional Gram spectrum are included once; the added ambient zero directions contribute nothing. $\square$

5.3 Energy and completion of the proof

Define $(\mathcal{T}h)(u) = h(u) + \int_I |u-v| h(v) dv$. For $f_0 = \cos(\sqrt{2}u)/Z_0$, $(\mathcal{T}f_0)'' = 0$ and evenness makes $\mathcal{T}f_0$ constant. Therefore a zero-mass perturbation has zero cross energy with f_0 . Orthogonality of the integer cosines, and $\mathcal{T} \cos(2\pi ju) = (1 - 1/(2\pi^2 j^2)) \cos(2\pi ju) + \text{constant}$, give

$$C(f) = \frac{1}{2} + \beta \cot \beta + \frac{\sum_{j=1}^{12} c_j^2 (1/2 - 1/(4\pi^2 j^2))}{2 \sin^2 \beta}. \quad (15)$$

The exact rational computation in Appendix A establishes

$$2 - C(f) > \frac{67216841}{10^8}. \quad (16)$$

In particular the energy cost of the perturbation is paid; the unperturbed energy constant is not substituted.

Proof of Theorem 1.1. The simple real positions lie in $(0, X_T]$, with $X_T = T \log T / (2\pi)$ and $X_T/N(T) \rightarrow 1$ by Riemann–von Mangoldt. Apply Proposition 4.1 to their Gram and then Lemma 5.1:

$$(1 - c + 42d_\varepsilon)n \geq 2N(T) - \text{tr } A_T^2 - BX_T - 7c.$$

Fix sufficiently small ε , let $T \rightarrow \infty$ using (13), and only then let $\varepsilon \rightarrow 0$. The $L^1 \cap L^2$ convergence implies $C(f_\varepsilon) \rightarrow C(f)$ in (12). Consequently

$$\liminf_T \frac{n}{N(T)} \geq \frac{2 - C(f) - B}{1 - c} > \frac{67216841 - 404350}{10^8 - 805003} = \frac{66812491}{99194997}.$$

The weak inequality stated in the theorem is therefore valid. The boundary charge $7c$ disappears only after division by $N(T)$. $\square$

6 Verification scope and further improvement

Input 2.1 is pinned to the public `dani-bound-2026` source [4] at commit

`d272437e7ebeb17b87c8c3d7cece592aa23785ab`.

Its raw `Solution.lean` is 1,675,641 bytes with SHA-256

`012c6ac5f9282158a686500dc0c967bf0a9b00e1a6192734d8f237bb23dd2d5f`.

The repository admission combines a continuous interval-soundness audit with all 240 exact integer compiled evaluations, seven complete finite structural checks, and Lean kernel-checked generic structural and minorant bridges. It includes the value and derivative tables, point lists, convexity chains, noncompact gap coverage and all 32 root trees. The evidence is recorded in repository note 483 and its linked audits [6]. The compiled evaluations are not 240 kernel-reduced proofs, and the full upstream zeta formalization was not rebuilt. The present result has that explicit finite-computation trust scope.

The new majorant and energy checker uses only Python’s standard-library arbitrary-precision integers and rational arithmetic, with assertions enabled. It does not run the much larger native search supporting the numerical headline in [5]. In particular that search’s `native_decide` computation axiom is not an input to this paper. Neither this checker nor the earlier symbolic bridges constitute a complete Lean verification of Theorem 1.1. The mathematical transfer, finite computation and formalized sublemmas are separate evidence. The manuscript was

prepared with AI assistance and internal independent reviews recorded in the repository; these reviews are distinct from external peer review.

The mechanism suggests a concrete next problem. Keep the same window and majorant, but seek a separated seven-gap certificate with reward x , mean gap fee 7α and a bounded telescoping storage correction. Its actual objective is $(2 - C(f) - 7\alpha)/(1 - x)$, not reward x alone. All storage endpoint terms must be retained before taking the limit. No such improved certificate is established here. The present proportion result also does not improve the project's separate zero-free-region bounds.

A Small exact rational certificate

The accompanying standard-library program is `scripts/am_lossless_majorant_certificate.py`, with report output `am-lossless-majorant-certificate.json`. From the repository root run

```
python -B scripts/am_lossless_majorant_certificate.py -check.
```

The checker refuses optimized Python execution. The following specifies the full continuous bridge independently of floating-point output.

For a nonnegative rational interval $q \subset [0, 10)$, it encloses $\cos \sqrt{q}$ and $\sin \sqrt{q}/\sqrt{q}$ by interval evaluation of

$$\sum_{k=0}^{12} \frac{(-1)^k q^k}{(2k)!}, \quad \sum_{k=0}^{12} \frac{(-1)^k q^k}{(2k+1)!},$$

with symmetric remainder bounds $q_{\max}^{13}/26!$ and $q_{\max}^{13}/27!$, respectively. The alternating tails decrease from the first omitted term. For the integer cosine modes the rational phase $2ju$ is reduced modulo two to $[-1, 1]$ before multiplication by π . No division by a small trigonometric argument is required. The normalization $Z_0 = \sin \beta/\beta$ has the exact argument squared $q = 1/2$.

Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, with the first 24 terms of each alternating arctangent series and its next-term remainder, certifies

$$3141592653/10^9 < \pi < 3141592654/10^9.$$

These enclosures, at all 129 nodes $u = k/256$, certify the strict lower margin $g(u) - f(u) > 9/100$. On all of I , $|\text{sinc}'| \leq \pi/2$ follows from its integral representation. Thus

$$|(g - f)'| \leq 4\gamma\pi\theta + \frac{\sqrt{2} + 2\pi \sum_j j|c_j|}{Z_0} < 4\gamma\frac{22}{7}\theta + \frac{3/2 + (44/7) \sum_j j|c_j|}{9/10} < 16.$$

This proves the global margin $47/800$ stated above.

For (16), use the same Taylor intervals to bound $\cos \beta/Z_0$ from above, $Z_0^2 = 2 \sin^2 \beta$ from below, and π from above in the positive correction in (15). All resulting comparisons are exact rational inequalities. A shorter independent certificate uses

$$\begin{aligned} u = \frac{3329448031}{4379443200} &> \cos \beta, & z = \frac{153778383027779}{167382319104000} &< Z_0, \\ d = \frac{180493}{213840} &< 2 \sin^2 \beta, & p = \frac{5277328977275528}{1679825970703125} &> \pi. \end{aligned}$$

These arise, respectively, from cosine terms through index 6, sinc terms through index 7, cosine at $\sqrt{2}$ through index 6, and Machin arctangent upper/lower sums through indices 4 and 1. Direct rational arithmetic gives

$$\frac{3}{2} - \frac{u}{z} - \frac{\sum_j c_j^2 (1/2 - 1/(4p^2 j^2))}{d} > \frac{67216841}{10^8}.$$

The positive margin is greater than $5.6 \cdot 10^{-10}$. The checker also verifies every span capacity, the close-pair constants, the majorant mass bound and (2). These small checks do not substitute for the full seven-dimensional certificate in Input 2.1.

References

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- [3] Y. Lamzouri, *A new proof that more than $2/3$ of the zeros of the Riemann zeta function are simple and on the critical line*, arXiv:2609.02882v1, Section 3, fixed-smoothing removal of the correlation weight.
- [4] *AM eight-point PC8CL certificate*, public dani-bound-2026 submission, fixed commit d272437e7eb17b87c8c3d7cece592aa23785ab, public proof source. The cosine-perturbed window and local weights are prior inputs.
- [5] K. M. Knausgård, *More than 83.9% of the zeros of the Riemann zeta function are distinct and more than 67.35% are simple and on the critical line*, arXiv:2610.08965v1, October 6, 2026, especially Section 5.
- [6] RH–Weil research project, public repository, note 483, the PC8 finite replay and independent continuous audits; note 498 and the present majorant certificate.